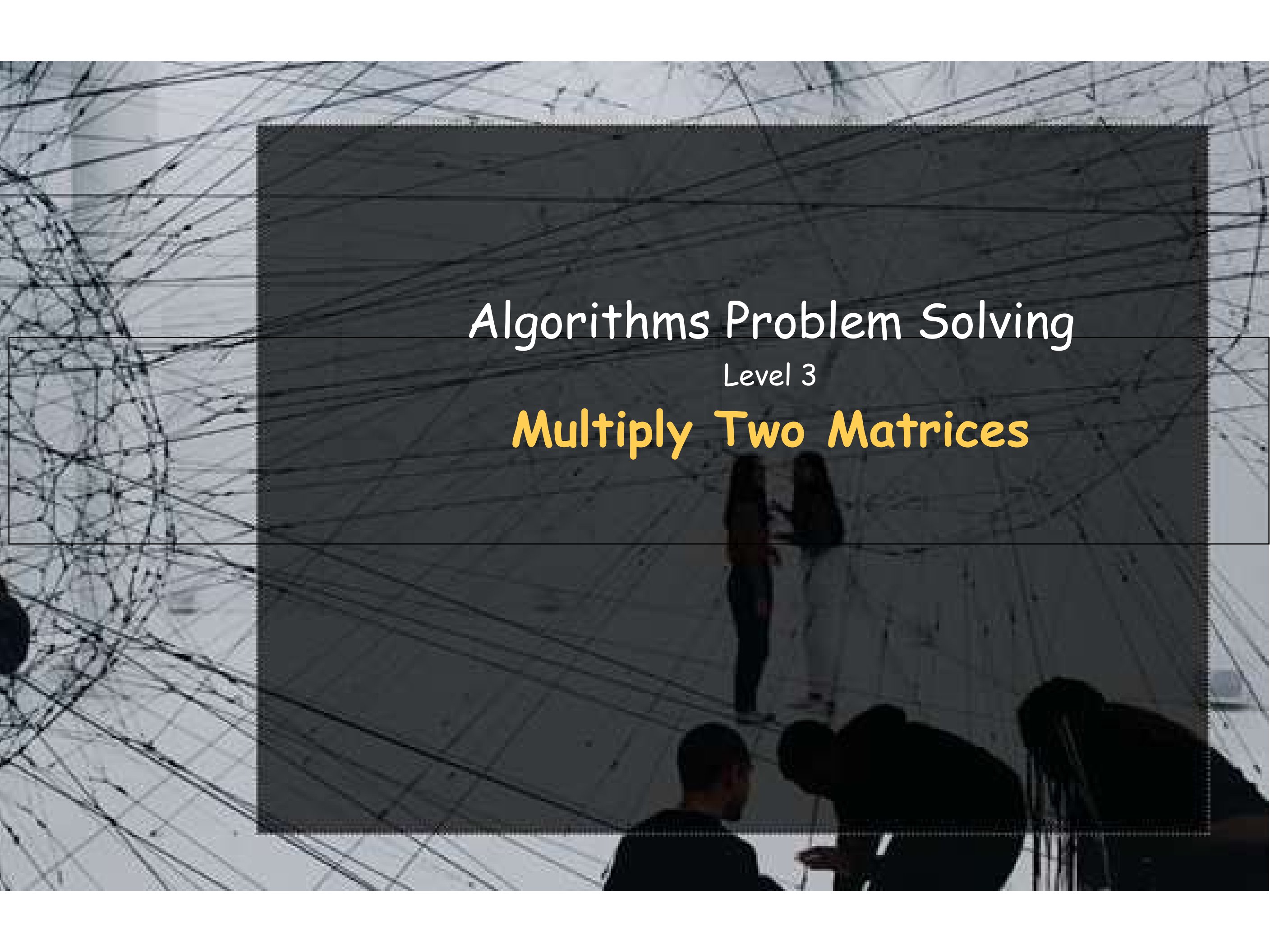


Algorithms Problem Solving

Level 3

Multiply Two Matrices



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Definition

Matrix Multiplication

 Program creates two 3×3 random matrices, multiplies them together, and stores the product in a third matrix.

Explanation

 Mathematical Product

 Formula: $\text{Result}[i][j] = \text{sum over } k \text{ of } (\text{Mat1}[i][k] * \text{Mat2}[k][j])$. Requires triple nested loops.

Procedures

1. Define FillMatrixWithRandomNumber() using range 1 to 10.
2. Define PrintMatrix() using setw(3).
3. Implement multiplyMatrices() with triple nested loops.
4. Execute multiplication in main() and print output.

💡 Example

1. Define FillMatrixWithRandomNumber() using range 1 to 10.
2. Define PrintMatrix() using setw(3).
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💡 Example

Matrix1:

08 07 07

01 06 10

10 01 09

Matrix2:

04 05 06

04 05 08

06 08 01

➡ Result

32 35 42

04 30 80

60 08 09

Note

- ◆ Seed random generator with `srand(time(0))`.

⊘ Warning

- ◆ Matrix multiplication requires the inner dimensions to match (for 3×3 matrices, dimensions are always compatible).
- ◆ Triple nested loops increase time complexity to $O(N^3)$.
- ◆ Uninitialized result matrix elements must be set to 0 before accumulation.
- ◆ Large element values can cause integer overflow during multiplication and accumulation.



Comparison

▨ `multiplyMatrices()`

- ✓ Multiplies two matrices using triple nested loops.
- ✓ Computes the dot product of rows and columns.
- ✓ Stores the result in a new 3×3 matrix.

▤ `ElementWiseMultiplication()`

- ✓ Multiplies corresponding elements directly (`Mat1[i][j] \* Mat2[i][j]`).
- ✓ Uses double nested loops instead of triple loops.
- ✓ Produces Hadamard product rather than matrix multiplication.

✦✦ Summary

- ✓ Implements canonical matrix multiplication using a classic triple-nested iteration loop structure.
- ✓ Computes dot products between rows of the first matrix and columns of the second matrix.
- ✓ Stores product calculations securely inside a newly initialized target result matrix.

Thank you 😊